

Development Phase – Team Role Allocation

To ensure efficient development and parallel progress, the project responsibilities are divided among four team members.

Member 1: Guru Charan – Frontend Developer

Responsibilities:

- Develop the user interface using React.
- Design responsive pages for:
 - Login/Register
 - User Dashboard
 - Tool Listings
 - Tool Details
 - Reservation History
 - Owner Dashboard
 - Admin Dashboard
- Integrate frontend with backend APIs.
- Handle client-side validation and navigation.

Primary Technologies:

- React
 - HTML/CSS
 - JavaScript
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Member 2: Vijaya Sai – Backend Developer

Responsibilities:

- Develop REST APIs using Node.js and Express.
- Implement Session authentication and authorization.
- Build business logic for:
 - User Management
 - Tool Management
 - Reservation Management
 - Reviews
 - Notifications
- Handle API security and error handling.

Primary Technologies:

- Node.js
 - Express.js
 - FastAPI
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Member 3: Harish (w. Vijay) – Database & Storage Developer

Responsibilities:

- Design and implement the relational database.
- Create tables, relationships, and constraints.
- Develop SQL queries and optimize database performance.
- Manage image storage (Cloudinary or local storage).
- Ensure data consistency and integrity.

Primary Technologies:

- PostgreSQL (or MySQL)
 - SQL
 - Cloudinary (optional)
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Member 4: Mohit Yash – Integration, Testing & Documentation Lead

Responsibilities:

- Integrate frontend, backend, and database modules.
- Perform unit, integration, and system testing.
- Identify and resolve bugs.
- Prepare UML diagrams, Software Architecture documentation, SRS, and the final project report.
- Coordinate deployment and final project presentation.

Primary Responsibilities:

- System Integration
- Testing
- Documentation
- Deployment
- Presentation